Mathematics Division, School of Advanced Sciences and Languages
VIT Bhopal University
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MAT2005-Transform Techniques and Difference Equations
Practice Set-1

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1. The fundamental period is the smallest positive period. Find it for

   (a). $\cos x,\ \sin x,\ \cos 2x,\ \sin 2x,\ \cos \pi x,\ \sin \pi x,\ \cos 2\pi x,\ \sin 2\pi x.$

   (b). $\cos nx,\ \sin nx,\ \cos \dfrac{2\pi x}{k},\ \sin \dfrac{2\pi x}{k},\ \cos \dfrac{2\pi nx}{k},\ \sin \dfrac{2\pi nx}{k}.$

2. The function $f(x) = x^3 - \pi^2 x$ has the Fourier series

$$f(x) = \sum_{n=1}^{\infty} \frac{12(-1)^n}{n^3} \sin nx.$$

   What is the value of $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^6}$? Ans: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}$

3. If

$$f(x) = \begin{cases} 0, & 0 < x < \frac{\pi}{2}, \\ \pi - x, & \frac{\pi}{2} < x < \pi, \end{cases}$$

   then show that

   - $f(x) = \dfrac{4}{\pi}\left[\dfrac{1}{1^2}\sin(x) - \dfrac{1}{3^2}\sin(3x) + \dfrac{1}{5^2}\sin(5x) - ...\right],$
   - $f(x) = \dfrac{\pi}{4} - \dfrac{2}{\pi}\left[\dfrac{1}{1^2}\cos(x) + \dfrac{1}{3^2}\cos(6x) + \dfrac{1}{5^2}\cos(10x) + \cdots\right].$

4. State and explain the Fourier Convergence Theorem with an example.

5. Find the Fourier series expansion of the function $\dfrac{\pi^2}{12} - \dfrac{x^2}{4}$; $-\pi < x < \pi$.

   Ans: $f(x) = \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \cos nx$

6. Show that in the interval $(0, 1)$

$$\cos \pi x = \frac{8}{\pi} \sum_{1}^{\infty} \left(\frac{n}{4n^2 - 1}\right) \sin 2n\pi x.$$

7. Find the Fourier series expansion of the function $f(x) = \frac{\pi - x}{2}$ in $0 < x < 2\pi$ and also deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + ...$

   Ans: The Fourier expansion of the given function on $[0, 2\pi]$ is given by

$$f(x) = \sum_{n=1}^{\infty} \frac{\sin(nx)}{n}.$$

8. Expand in Fourier series of periodicity $2\pi f(x) = x\sin x$, for $0 < x < 2\pi$.

Ans: The Fourier expansion of the given function on $[0, 2\pi]$ is given by

$$x\sin x = 1 - \frac{1}{2}\cos x + 2\sum_{n=2}^{\infty}\frac{\cos(nx)}{n^2-1} + \pi\sin x.$$

9. Find the Fourier series expansion of the periodic function

$$f(x) = x, -\pi \le x \le \pi, f(x+2\pi) = f(x)$$

Ans: The Fourier expansion of the given function on $[-\pi, \pi]$ is given by

$$2\left[\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \cdots\cdots\right].$$

10. Find the Fourier series of $f(x)$ with period $2l$, defined by $f(x) = \begin{cases} l - x, & 0 < x < l \\ 0, & l < x < 2l \end{cases}$.

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$.

Ans:

$$f(x) = \frac{l}{4} + \frac{2l}{\pi^2}\sum_{n=1,3,\ldots}^{\infty}\frac{1}{n^2}\cos\frac{n\pi x}{l} + \frac{l}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi x}{l}$$

11. Determine whether the function $f(x) = \begin{cases} \frac{1}{2} + x, & \text{when } -1 \le x \le 0, \\ \frac{1}{2} - x, & \text{when } 0 \le x \le 1, \end{cases}$ is an even function. If so, find its Fourier series.

Ans:$f(x) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty}\frac{4}{(n\pi)^2}\cos(n\pi x)$**.**

12. Find the Fourier series representation of $f(t)$ in $(0, 2T)$ of the half wave rectifier defined by

$$f(t) = \begin{cases} E\sin\frac{\pi t}{T}, & 0 < t < T, \\ 0, & \text{otherwise.} \end{cases}$$

Ans:$f(t) = \frac{E}{\pi} + \sum_{n=2,4,6,\cdots}^{\infty}\frac{2E}{\pi(n^2-1)}\cos\frac{n\pi t}{T} + \frac{E}{2}\sin\frac{\pi t}{T}$

13. Expand $f(x) = \sqrt{1-\cos x}, \quad 0 < x < 2\pi$, in a Fourier series. Hence evaluate $\frac{1}{1\times 3} + \frac{1}{3\times 5} + \frac{1}{5\times 7} + \cdots$

Ans:$f(x) = \sqrt{1-\cos x} = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi}\left[\frac{1}{1\times 3}\cos x + \frac{1}{3\times 5}\cos 2x + \frac{1}{5\times 7}\cos 3x + \cdots\right]$

$$\frac{1}{2} = \frac{1}{1\times 3} + \frac{1}{3\times 5} + \frac{1}{5\times 7} + \cdots$$

14. Find the Fourier series expansion for the function $f(x) = x^3, \ -\pi < x < \pi$.

Ans: $f(x) = 2\sum_{n=1}^{\infty}\left(\frac{6}{n^3} - \frac{\pi^2}{n}\right)(-1)^n\sin nx.$

15. Find the Fourier half range series for the function $f(x) = (2x-1);\ \ 0 < x < 1$.

Ans: $f(x) = -\dfrac{2}{\pi}\left[\sin 2\pi x + \dfrac{1}{2}\sin 4\pi x + \dfrac{1}{3}\sin 6\pi x \cdots\right]$.

16. Find the Fourier Series for $f(x) = |sin\ x|$, $-\pi < x < \pi$, and hence show that $\dfrac{1}{1.3} + \dfrac{1}{3.5} + \dfrac{1}{5.7} + \dfrac{1}{7.9} + \cdots = \dfrac{1}{2}$.

Ans: $f(x) = |sin\ x| = \dfrac{2}{\pi} - \dfrac{4}{\pi}\left[\dfrac{1}{1\cdot 3}\cos\ 2x + \dfrac{1}{3\cdot 5}\cos\ 4x + \dfrac{1}{5\cdot 7}\cos\ 6x + \cdots\right]$

17. Is the given function even or odd or neither even nor odd? Find its Fourier series. Show details of your work. $f(x) = x|x|$ and $f(x+2) = f(x)$ on $(-1, 1)$.

Ans: $f(x) = \displaystyle\sum_{n=1}^{\infty}\left(\frac{-4+4(-1)^n}{n^3\pi^3} - \frac{2(-1)^n}{n\pi}\right)\sin n\pi x$

18. Write down the Dirichlet condition of convergence of the Fourier series. Is it a sufficient condition for a Fourier series to converge at the value of function? Justify. Give an example of a function with a finite number of discontinuities but an infinite number of maxima and minima in $[-\pi, \pi]$.

Ans: No. The Fourier series of $f(x) = \begin{cases} x, & \textbf{if}\ \ x \in [-\pi, 0) \\ x+5, & \textbf{if}\ \ x \in [0, \pi] \end{cases}$ **does not converge to the value of the function at** $x = 0$**. Take the function** $f(x) = \begin{cases} \sin\frac{1}{x}, & \textbf{if}\ \ x \in [-\pi, \pi] \setminus \{0\} \\ 0, & \textbf{if}\ \ x = 0 \end{cases}$

19. Find the Fourier Series for the function for which the graph is given

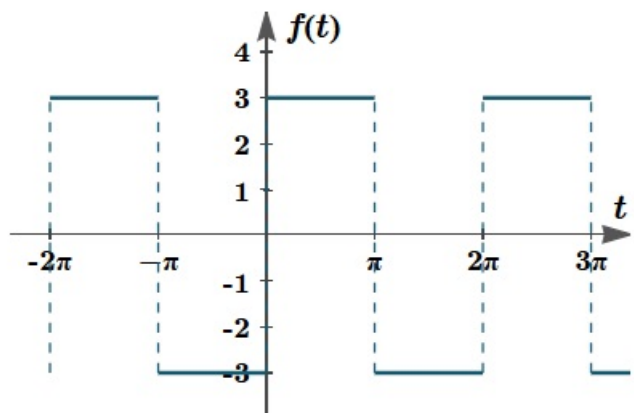

Figure 1: Graph of an odd periodic square wave function.

Ans: $f(x) = \begin{cases} -3, & -\pi < t < 0 \\ 3, & 0 < t < \pi \end{cases}$ **and:** $f(x) = \sum_{n=1}^{\infty} \frac{12}{\pi}\frac{\sin(2n-1)t}{(2n-1)}$

20. Obtain the Fourier series of the function

$$f(x) = \begin{cases} 0, & -\pi \le x \le 0, \\ \frac{1}{4}\pi x, & 0 \le x \le \pi \end{cases}$$

Ans: $f(x) = \dfrac{\pi^2}{6} + \displaystyle\sum_{n=1}^{\infty}\frac{[(-1)^n - 1]}{4n^2}\cos nx - \frac{(-1)^n}{4n}\sin nx$**.**

21. Compute the Fourier series of each function below (assume the function is 2π-periodic).

(a) $f(x) = \begin{cases} 0 & -\pi < x < -\pi/2 \\ 1 & -\pi/2 < x < \pi/2 \\ 0 & \pi/2 < x < \pi \end{cases}$ **Ans:** $f(x) = 1 + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin \frac{n\pi}{2} \cos(nx)$

(b) $f(x) = \begin{cases} \pi + x & -\pi < x < 0 \\ \pi - x & 0 < x < \pi \end{cases}$

22. Obtain the Sine series of the function

$$f(x) = \begin{cases} cx, & 0 \le x \le \frac{l}{2}, \\ c(l-x), & \frac{l}{2} \le x \le l \end{cases}$$

Ans: $f(x) = \dfrac{4cl}{\pi^2} \displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{2} \sin \frac{n\pi x}{l}$.

23. Obtain the Fourier Sine series of the function

$$f(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi, \end{cases}$$

where $f(x + 2\pi) = f(x)$.

Ans: $\frac{4}{\pi}\left[\frac{1}{1}\sin(x) + \frac{1}{3}\sin(3x) + \frac{1}{5}\sin(5x) + \frac{1}{7}\sin(7x) + \cdots\right]$

24. Find the Fourier half range sine series for the function $f(x) = e^x$, $0 < x < 1$. Ans: $f(x) = 2\pi \sum_{n=1}^{\infty} \frac{n(1-(-1)^n e)}{1+n^2\pi^2} \sin n\pi x$

25. Obtain half range sine and cosine series for the function

$$f(x) = \begin{cases} x, & 0 < x < \frac{\pi}{2}, \\ (\pi - x), & \frac{\pi}{2} < x < \pi \end{cases}$$

Ans: sine series $f(x) = \displaystyle\sum_{n=1}^{\infty} \frac{4}{\pi n^2} \sin \frac{n\pi}{2} \sin nx$

cosine series $f(x) = \dfrac{\pi}{4} + \dfrac{2}{\pi} \displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} \left(2\cos \frac{n\pi}{2} - 1 - (-1)^n\right) \cos nx$

26. Find the Fourier cosine and sine series of the function $f(x) = 1, 0 \le x \le 2$.

Ans: The Fourier cosine series is $f(x) = 1$. The Fourier sine series is

$$f(x) = \frac{4}{\pi}\left[\sin\left(\frac{\pi x}{2}\right) + \frac{1}{3}\sin\left(\frac{3\pi x}{2}\right) + \frac{1}{5}\sin\left(\frac{5\pi x}{2}\right) + \cdots\right].$$

27. Develop the half-range cosine series expansion of $f(x) = x(2-x)$ in $0 \le x \le 2$, and hence find the sum of the series $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$.

Ans: $1 - \dfrac{1}{2^2} + \dfrac{1}{3^2} - \dfrac{1}{4^2} + \cdots = \dfrac{\pi^2}{12}$.

28. Find the half range Fourier cosine and sine series expansion of following functions $f(x) = e^{-x} \qquad 0 < x < \pi$

Ans:

$$\text{half range Fourier cosine Series} = \frac{1-e^{-\pi}}{\pi} - \frac{2}{\pi}\sum_{n=1}^{\infty}\left(\frac{1-(-1)^n e^{-\pi}}{n^2+1}\right)\cos nx$$

$$\text{half range Fourier cosine Series} = \frac{2}{\pi}\sum_{n=1}^{\infty} n\left(\frac{1-(-1)^n e^{-\pi}}{n^2+1}\right)\sin nx$$

29. Obtain the Fourier cosine series for $f(x) = x\sin x$, $x \in (0,\pi)$. Hence evaluate the value of $\frac{\pi-2}{4}$
Ans:$f(x) = 1 - \frac{1}{2}\cos x - \frac{2}{1.3}\cos 2x + \frac{2}{2.4}\cos 3x -, \ldots$ **and:** $\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7}, \ldots = \frac{\pi-2}{4}$

30. Expand $f(x) = \pi - x, 0 < x < \pi$ as a Fourier Cosine series. **Ans:** $f(x) = \frac{\pi}{2} + \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1}{n^2}\cos nx$

31. Expand $f(x) = \pi x - x^2, 0 < x < \pi$ as a Fourier Cosine series and hence use Parseval's identity to deduce. $\sum_{1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}$.

Ans: $f(x) = \frac{\pi^2}{6} - 4\sum_{n=2}^{\infty}\frac{1}{n^2}\cos nx$**,** n **is even**

32. Obtain the Fourier series for $f(x) = x^2$ in $-\pi < x < \pi$. Hence show that

$$\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots = \frac{\pi^2}{90}$$

Hint: From the Fourier series of $f(x) = x^2$**,**

$$a_0 = \frac{2\pi^2}{3},\ a_n = \frac{(-1)^n 4}{n^2},\ b_n = 0, \forall n.$$

Then use Parseval's identity

$$\frac{{a_0}^2}{2} + \sum_{n=1}^{\infty} {a_n}^2 + {b_n}^2 = \frac{1}{L}\int_{-L}^{L} f^2(x)\ dx.$$

33. Using Parseval's Identity, prove that for a periodic function $f(x)$ with period 2π, the sum of the squares of its Fourier coefficients equals the average value of the square of the function over one period.
Ans: Fourier series

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n\cos(nx) + b_n\sin(nx))$$

and

$$\frac{1}{\pi}\int_{-\pi}^{\pi}|f(x)|^2 dx = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}\left(a_n^2 + b_n^2\right)$$

34. Find the Fourier series for the function $f(x) = x^2, -\pi < x < \pi$ where $f(x) = f(x+2\pi)$. Using Parseval's equality show that

$$\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots = \frac{\pi^4}{90}$$

.

Ans: $x^2 = \frac{1}{3}\pi^2 + \sum_{n=1}^{\infty} \frac{4}{n^2}(-1)^n \cos\ nx$.

35. State and prove Parseval's identity of half-range Fourier cosine series for $f(x) = x^2$, $x \in (0, 1)$.

36. Find the Fourier series as far as the second harmonic to represent the function given by table below:

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°
f(x)	2.34	3.01	3.69	4.15	3.69	2.20	0.83	0.51	0.88	1.09	1.19	1.64

Ans: $f(x) = 2.1015 + 0.559\cos x - 0.519\cos 2x + \cdots + 1.535\sin x - 0.091\sin 2x + \cdots$

37. Expand the function $y = f(x)$ in the form of a Fourier series up to second harmonic from the following values of x and y.

x	0	2	4	6	8	10	12
y	9.0	18.2	24.4	27.8	27.5	22.0	9.0

Ans. $a_0 = 42.96$; $a_1 = -8.22$; $a_2 = -3.08$; $b_1 = -1.99$; $b_2 = -0.203$

38. The following data gives the variations of a periodic current over a period

t secs	0	$\frac{T}{6}$	$\frac{T}{3}$	$\frac{T}{2}$	$\frac{2T}{3}$	$\frac{5T}{6}$	T
A amps	9	18.2	24.4	27.8	27.5	22	9

Find numerically the direct current part of the variable current and obtain the amplitudes upto the second harmonic.

Ans: $A = 21.4835 + (-8.217\cos\theta - 1.9918\sin\theta) + (-3.083\cos 2\theta - 0.202\sin 2\theta)$
Amplitude of the first harmonic : 8.455, Amplitude of the second harmonic : 3.09

39. Identify the harmonics of the following signal

$$X(t) = 2\sin(3t + 15°) + 6\sin(9t + 30°) + 8\sin(12t + 15°).$$

Hint:

Fundamental frequency $= 3$
1st harmonic $= 2\sin(3t + 15°)$
3rd harmonic $= 6\sin(9t + 30°)$
4th harmonic $= 8\sin(12t + 15°)$

40. Determine the frequency spectrum of the rectangular waveform with $a_0 = 1$, $a_n = 0$ and $b_n = \dfrac{1 - \cos n\pi}{n\pi}$.

Ans:

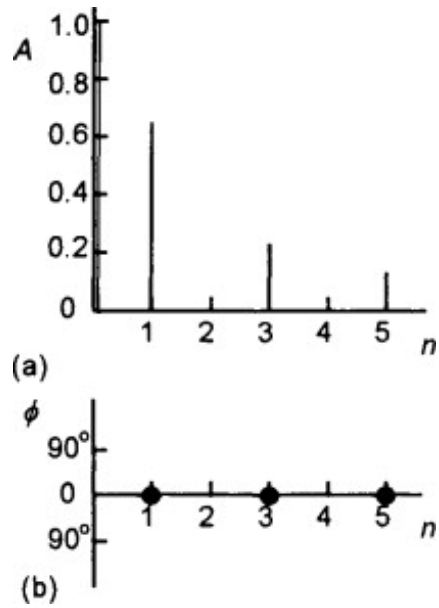

Figure 2: Frequency Spectrum